

# **Consumer Preferences and Choice**

Lecture 1

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# Introduction

# Why Study Microeconomics?

- **Non-economists typically think of economics when it concerns the *macro stuff*** (unemployment, inflation, growth, recessions etc.)
- **However, the macro economy is made up of millions of *micro* decisions.**
  - **People:** what to buy, how much to work, how much to save
  - **Firms:** pricing, hiring, location, investment
  - **Government:** infrastructure, regulation, tax policy
- **Markets coordinate these decisions invisibly and *often* effectively.**
  - Gas is at the pump when you need it
  - Jobs exist for qualified workers
  - Products get delivered

# Why Study Microeconomics?

- **However, sometimes markets produce undesirable outcomes or fail.**
  - Each micro failure may seem small (one shortage, one person unemployed, one overpriced good)
  - But small failures add up to large macro consequences
- **In this course, we will study:**
  - How individuals and firms make decisions?
  - How markets coordinate these decisions?
  - When markets work efficiently and when they fail?
  - What interventions can help?

# Course Roadmap

## Competitive Markets

- Consumer choice (1-2)
- Firm production (3)
- Market equilibrium (4)
- Efficiency and welfare (5)

## Market Power

- Monopoly and oligopoly (6-7)
- Labor markets and monopsony (8)

## Market Failures

- Information asymmetries (9)
- Externalities and public goods (10)

## Uncertainty & Strategy

- Decisions under uncertainty (11)
- Game theory (12)

# Today's Lecture

**Central question:** How do individuals make choices?

## Economic Approach to Choice

- Individuals have **preferences** over outcomes
- These preferences are **rational** (satisfy certain axioms)
- We can represent preferences with a **utility function**
- Given constraints, individuals **maximize utility**

## Does everyone really “maximize utility”?

- Maybe not consciously—but people behave “as if” they do
- The model predicts behavior relatively well
- Alternative: behavioral economics

# Preferences and Utility

# Preferences: Basic Setup

We consider a consumer choosing between bundles of goods.

- **Consumption bundle:**  $(x, y)$  where  $x, y \geq 0$
- **Preference relation:**  $\succeq$  (weakly preferred to)
  - $A \succeq B$ : Bundle  $A$  is at least as good as bundle  $B$
  - $A \succ B$ : Bundle  $A$  is strictly preferred to  $B$
  - $A \sim B$ : Consumer is indifferent between  $A$  and  $B$

**Key question:** What properties should preferences satisfy for them to be “rational”?



# Axioms of Rational Choice

## 1. Completeness

For any two bundles  $A$  and  $B$ , the consumer can state which is preferred or that they are indifferent:

$$A \succeq B, \quad B \succeq A, \quad \text{or both (indifference)}$$

**Interpretation:** Consumers can always make comparisons. Rules out indecision.

## 2. Transitivity

If  $A \succeq B$  and  $B \succeq C$ , then  $A \succeq C$

**Interpretation:** Preferences are internally consistent. No cycles.

# Axioms of Rational Choice (cont.)

## 3. Continuity

Small changes in consumption bundles lead to small changes in preferences.

**Technical:** For any bundle  $A$ , the sets  $\{B : B \succeq A\}$  and  $\{B : A \succeq B\}$  are closed.

**Interpretation:** No sudden jumps. Preferences are “smooth.”

## 4. Non-satiation (Monotonicity)

More is better: If  $A$  has at least as much of everything as  $B$ , and strictly more of at least one good, then  $A \succ B$ .

**Interpretation:** Consumers always prefer more to less (at least weakly).

# Axioms of Rational Choice (cont.)

## 5. Convexity

Averages are preferred to extremes. If  $A \sim B$ , then:

$$\lambda A + (1 - \lambda)B \succeq A \text{ for } \lambda \in [0, 1]$$

**Interpretation:** Consumers prefer balanced consumption bundles. Diminishing marginal rate of substitution.

**Example:** If you're indifferent between (6 apples, 0 oranges) and (0 apples, 6 oranges), you prefer (3 apples, 3 oranges) to either extreme.

# When Do Axioms Fail?

## Behavioral Economics Violations

- **Framing effects:** Preferences change based on how options are presented
- **Intransitivity:** Preference reversals in complex choices
- **Present bias:** Time-inconsistent preferences
- **Reference dependence:** Preferences depend on current endowment (loss aversion)
- **Bounded rationality**
  - Too many options → choice paralysis
  - Computational constraints
  - Limited attention

# From Preferences to Utility

**Key Theorem:** If preferences satisfy completeness, transitivity, continuity, and monotonicity, then there exists a continuous **utility function**  $U(x, y)$  that represents them:

$$A \succeq B \iff U(A) \geq U(B)$$

**Interpretation:** We can assign numbers to bundles such that higher numbers = more preferred.

**Important:** Utility is **ordinal**, not **cardinal**

- Only the **ranking** matters, the magnitude of utility has no meaning.
- $U(A) = 10, U(B) = 5$  tells us  $A \succ B$ . It does NOT mean “A is twice as good as B”

# Monotonic Transformations

- Since utility is ordinal, we can apply any **strictly increasing transformation** without changing preferences:
- If  $U(x, y)$  represents preferences, so does  $V(x, y) = f(U(x, y))$  for any strictly increasing  $f$ .
- **Examples:**
  - $U(x, y) = xy$  and  $V(x, y) = \ln(xy)$  represent the same preferences
  - $U(x, y) = x^{0.5}y^{0.5}$  and  $V(x, y) = xy$  represent the same preferences

**Why this matters:** We can transform utility functions to make calculations easier.

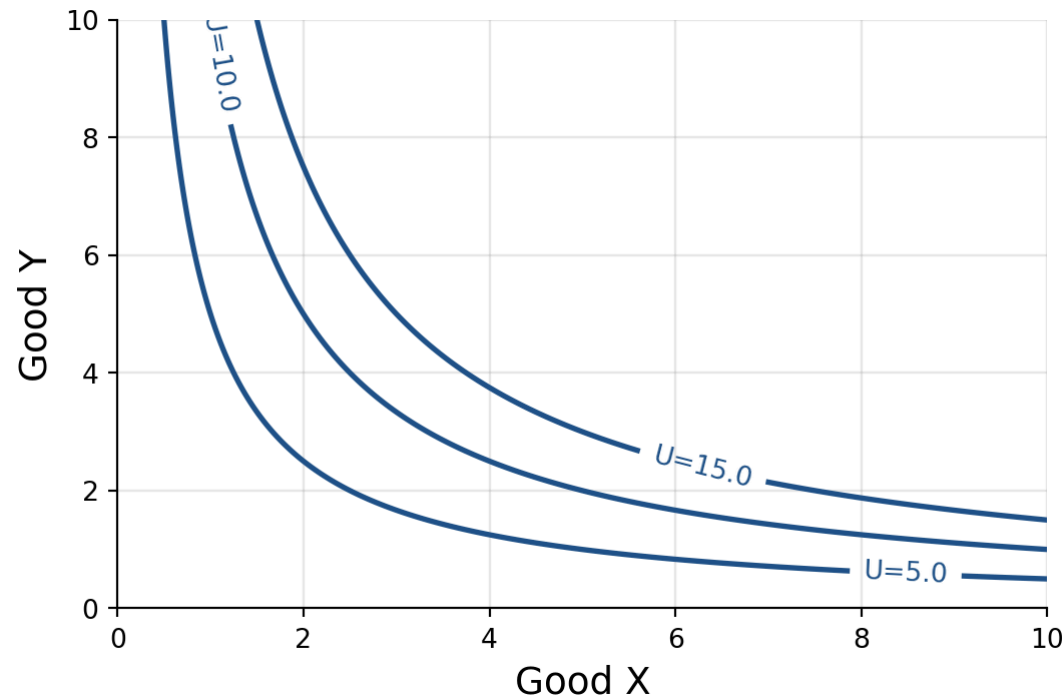
# **Indifference Curves & MRS**

# Indifference Curves

An **indifference curve** is the set of all bundles that give the same utility level:

$$IC(U_0) = \{(x, y) : U(x, y) = U_0\}$$

**Interpretation:** The consumer is indifferent between any two points on the same curve.



Indifference Curves for  $U(x, y) = xy$



# Properties of Indifference Curves

Under our axioms, indifference curves must be:

1. **Downward sloping** (from non-satiation)
  - To keep utility constant, if  $x$  increases,  $y$  must decrease
2. **Do not cross** (from transitivity)
  - If they crossed, we'd have  $A \sim B$  and  $A \sim C$  but  $B \not\sim C$
3. **Convex to the origin** (from convexity of preferences)
  - Averages preferred to extremes
  - Equivalently: diminishing marginal rate of substitution
4. **Higher curves represent higher utility** (from monotonicity)

# Marginal Rate of Substitution (MRS)

- The **marginal rate of substitution** is the rate at which the consumer is willing to trade good  $Y$  for good  $X$  while maintaining constant utility.
- **Geometrically**:  $MRS = -(\text{slope of indifference curve})$

$$MRS = -\frac{dy}{dx} \Big|_{U=\text{const}}$$

- **Interpretation**: How many units of  $Y$  are you willing to give up to get one more unit of  $X$ ?
- **Example**: If  $MRS = 2$ , you're willing to give up 2 units of  $Y$  to get 1 more unit of  $X$  (and remain indifferent).

# Deriving the MRS Formula

Along an indifference curve, utility is constant:  $U(x, y) = \bar{U}$ . Taking the total differential:

$$dU = \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy = 0$$

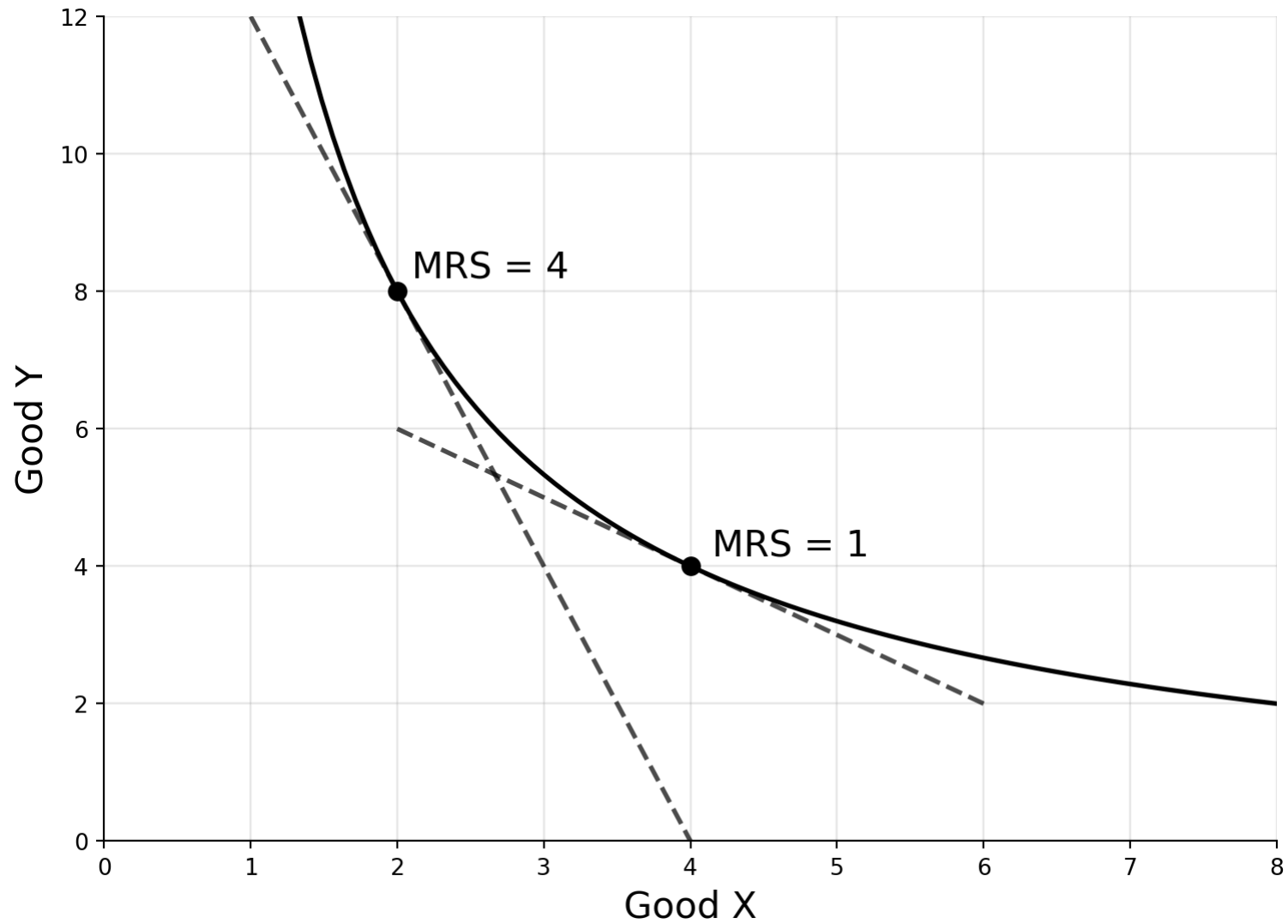
Rearranging:

$$\frac{\partial U}{\partial y} dy = -\frac{\partial U}{\partial x} dx \quad \rightarrow \quad \frac{dy}{dx} = -\frac{\partial U / \partial x}{\partial U / \partial y} = -\frac{MU_x}{MU_y}$$

Therefore:

$$\boxed{MRS = -\frac{dy}{dx} = \frac{MU_x}{MU_y}}$$

# Visualizing MRS



# Diminishing MRS

## Convexity assumption → Diminishing MRS

As you consume more of good  $X$  (moving right along an IC), the MRS decreases:

- When you have a lot of  $Y$  and little  $X$ : High MRS (willing to give up a lot of  $Y$  for more  $X$ )
- When you have a lot of  $X$  and little  $Y$ : Low MRS (not willing to give up much  $Y$  for more  $X$ )

**Economic intuition:** Scarcity increases value. The less you have of something, the more you value additional units.

# Common Utility Functions

# Perfect Substitutes

Goods that can be substituted at a constant rate.

$$U(x, y) = ax + by$$

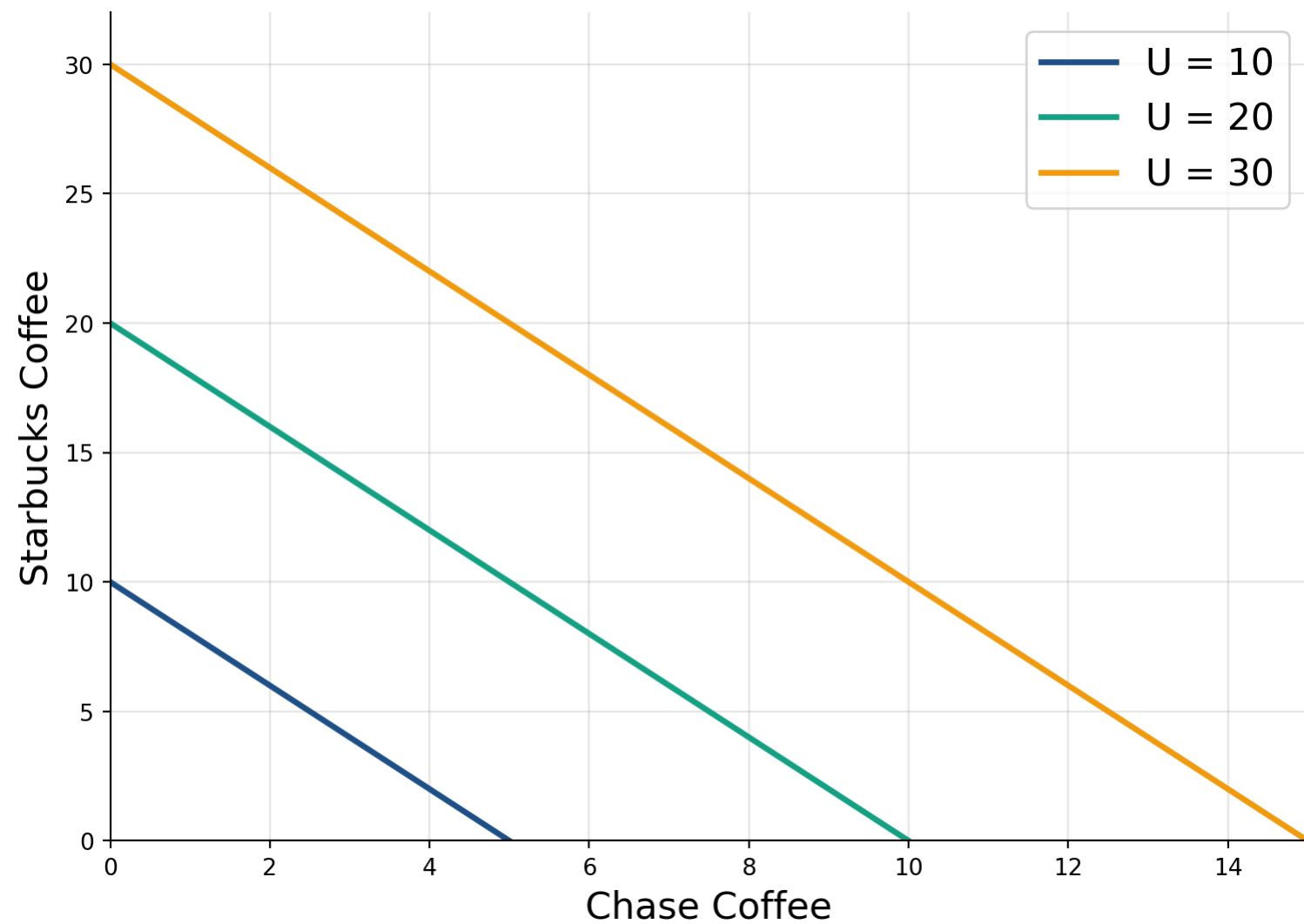
## Examples:

- Different brands of the same product (e.g., Coke vs Pepsi for some)
- Coffee from different cafes
- Generic vs brand-name drugs (if truly equivalent)

## Key features:

- Indifference curves are **straight lines**
- MRS is **constant**:  $MRS = a/b$
- Consumer willing to trade at fixed rate regardless of bundle

# Perfect Substitutes Graph



Perfect Substitutes:  $U = 2x + y$



# Perfect Complements

Goods that must be consumed in fixed proportions.

$$U(x, y) = \min\{ax, by\}$$

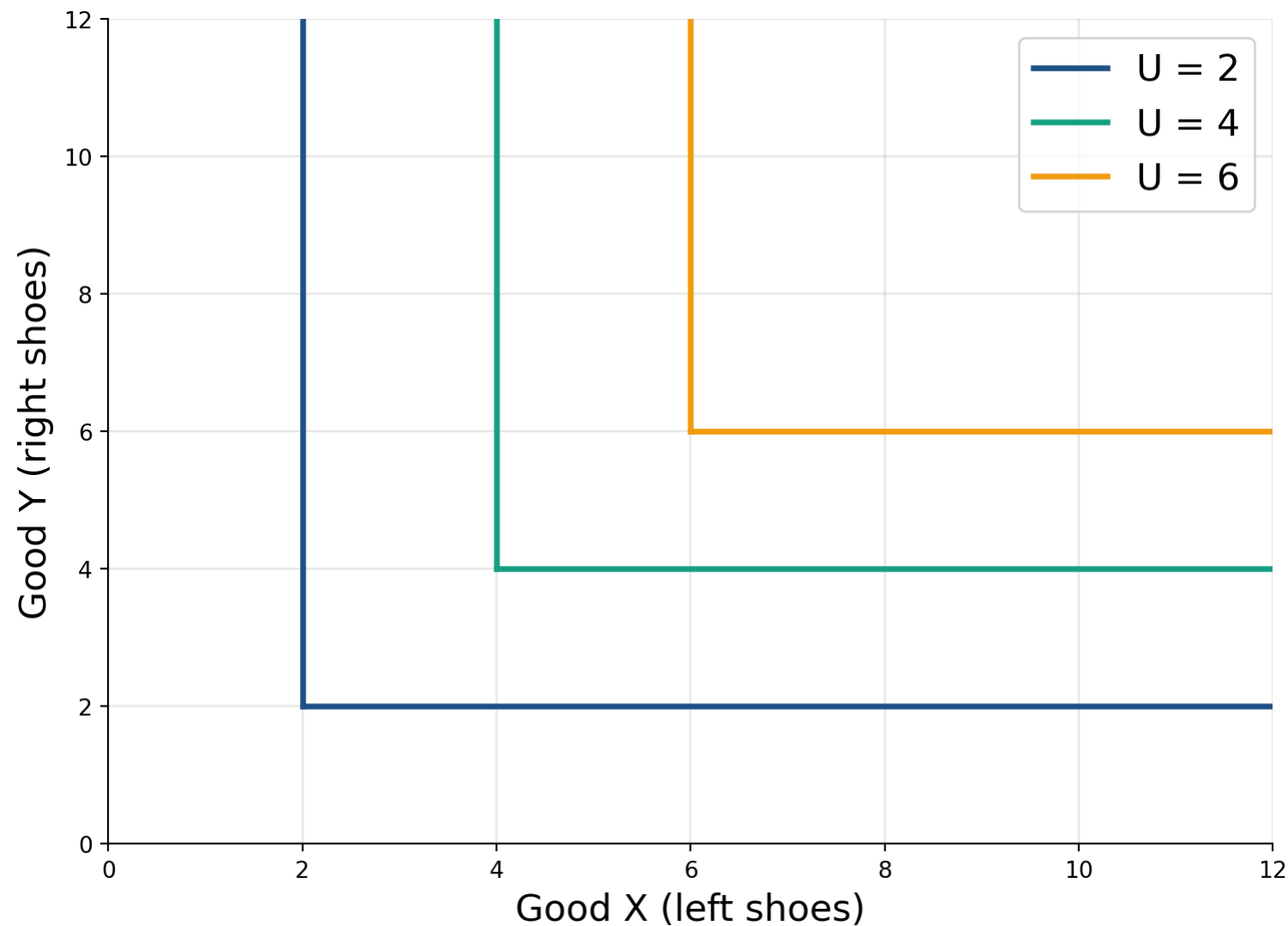
## Examples:

- Left and right shoes
- Cars and tires (need 4 tires per car)
- Computers and monitors

## Key features:

- Indifference curves are **L-shaped**
- Consumed in fixed ratio:  $x/y = b/a$
- MRS is undefined (technically infinite or zero, depending on side)

# Perfect Complements Graph



Perfect Complements:  $U = \min\{x, y\}$

# Cobb-Douglas Utility

The most widely used functional form in economics:

$$U(x, y) = x^{\alpha} y^{\beta}$$

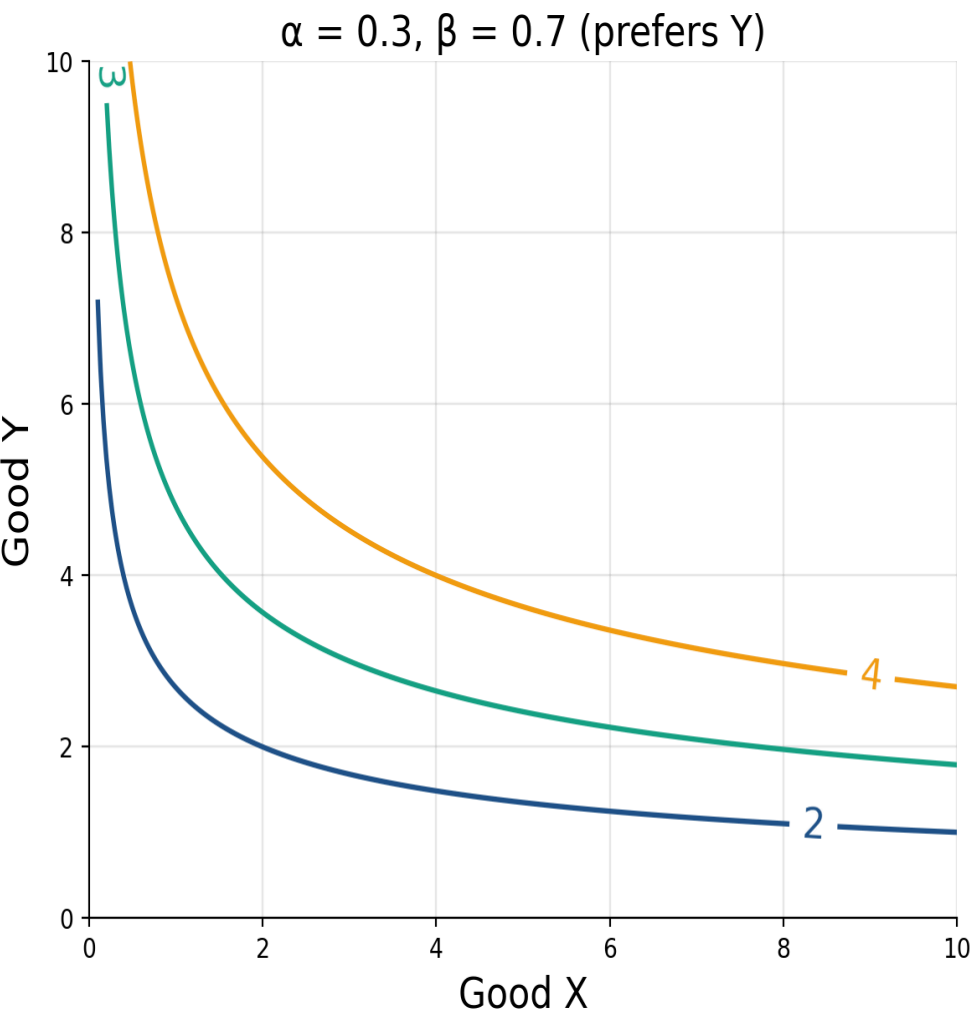
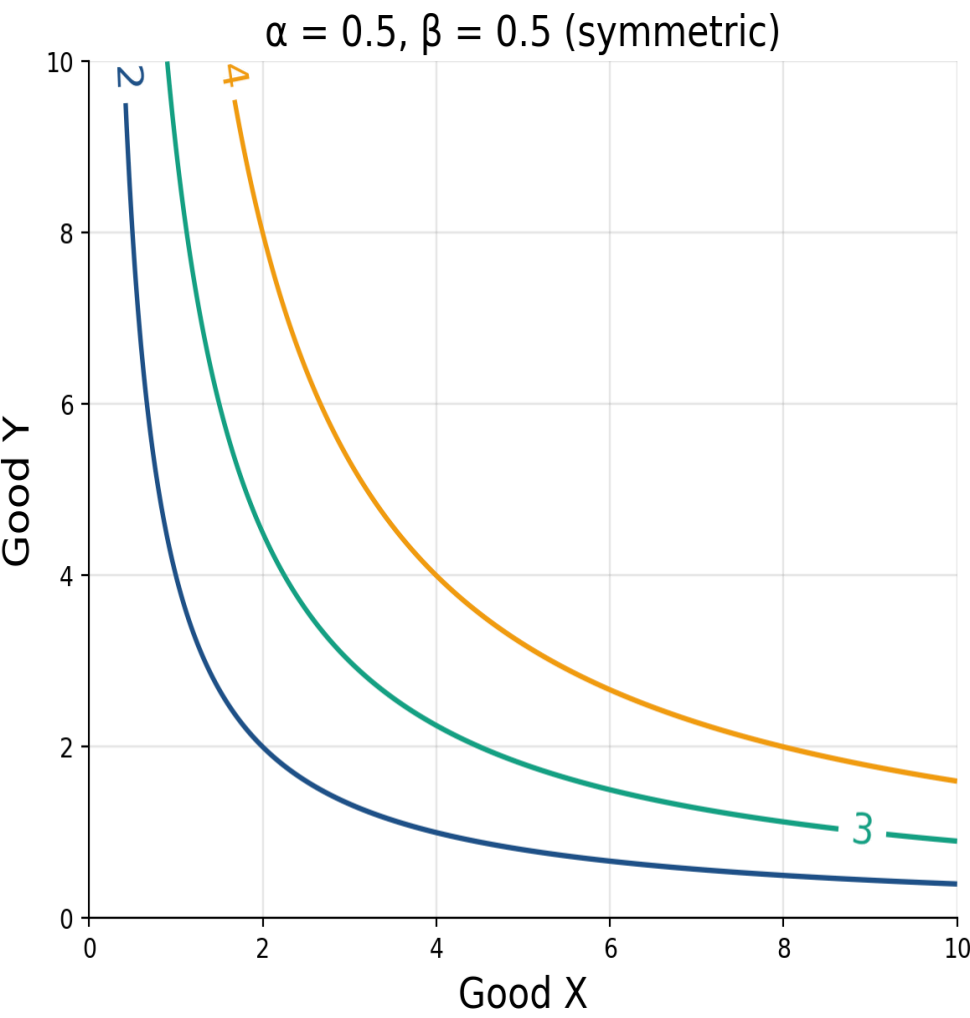
Or equivalently (applying monotonic transformation):

$$U(x, y) = \alpha \ln x + \beta \ln y$$

## Key features:

- Smooth, convex indifference curves
- Interior solutions (typically)
- Constant expenditure shares
- $MRS = \frac{\alpha}{\beta} \cdot \frac{y}{x}$

# Cobb-Douglas Graph



Cobb-Douglas Utility:  $U = x^\alpha y^\beta$

# Cobb-Douglas: MRS Calculation

For  $U(x, y) = x^\alpha y^\beta$ :

**Step 1:** Find marginal utilities

$$MU_x = \frac{\partial U}{\partial x} = \alpha x^{\alpha-1} y^\beta, \quad MU_y = \frac{\partial U}{\partial y} = \beta x^\alpha y^{\beta-1}$$

**Step 2:** Calculate MRS

$$MRS = \frac{MU_x}{MU_y} = \frac{\alpha x^{\alpha-1} y^\beta}{\beta x^\alpha y^{\beta-1}} = \frac{\alpha}{\beta} \cdot \frac{y}{x}$$

MRS depends on the ratio  $y/x$  and the preference parameters  $\alpha/\beta$ .

# CES Utility

Constant Elasticity of Substitution (CES) utility function:

$$U(x, y) = (ax^\rho + by^\rho)^{1/\rho}, \quad \rho \leq 1, \rho \neq 0$$

Elasticity of substitution:  $\sigma = \frac{1}{1-\rho}$

Special cases:

- $\rho \rightarrow -\infty$ : Perfect complements ( $\sigma = 0$ )
- $\rho = 0$ : Cobb-Douglas ( $\sigma = 1$ )
- $\rho = 1$ : Perfect substitutes ( $\sigma = \infty$ )

**Flexibility:** CES nests all three cases mentioned.

# Utility Maximization

# The Budget Constraint

Consumers have limited income  $I$  and face prices  $p_x, p_y$  for goods:

$$p_x \cdot x + p_y \cdot y \leq I$$

**Budget line:** Set of bundles that cost exactly  $I$ :

$$p_x \cdot x + p_y \cdot y = I$$

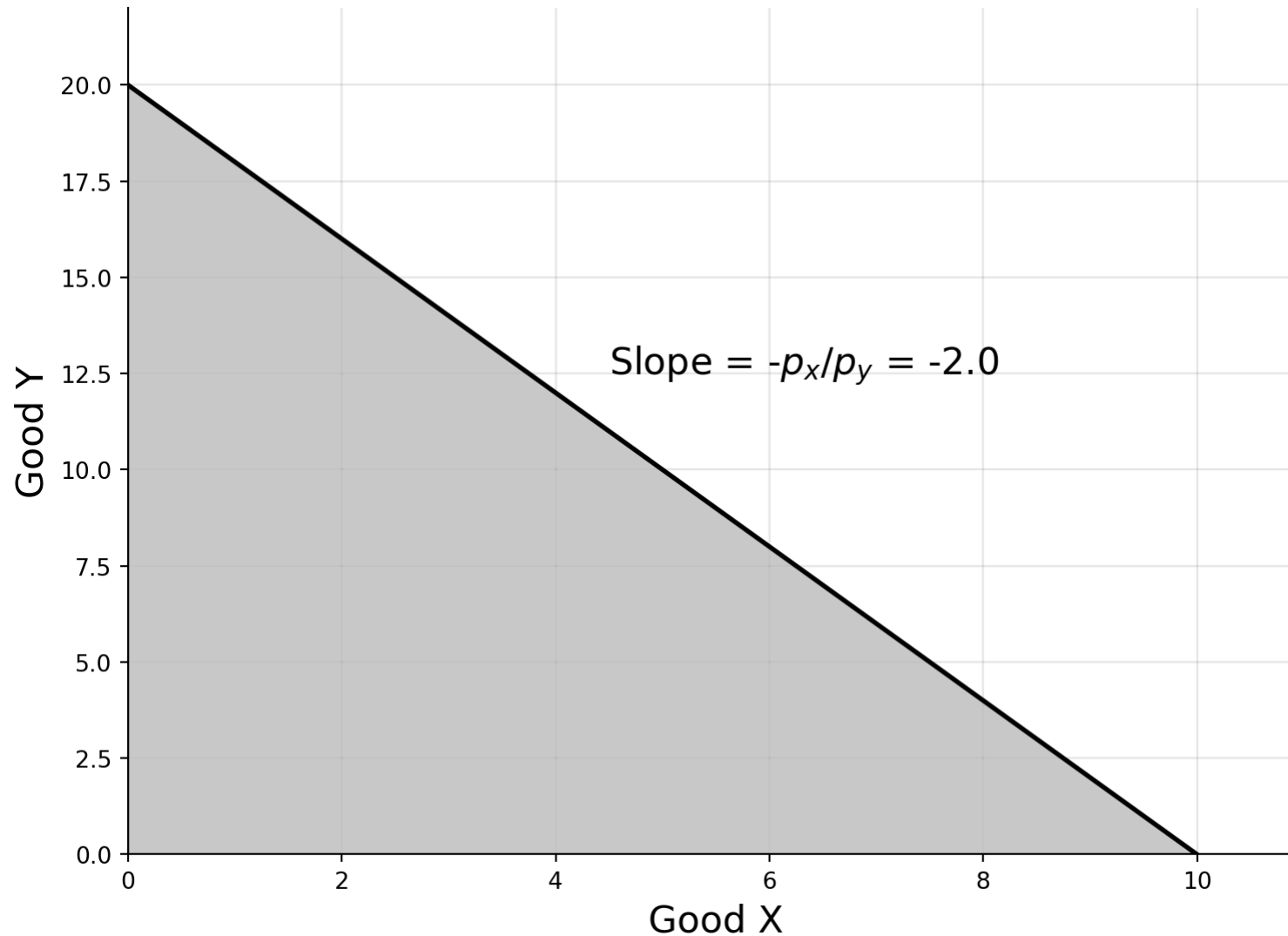
Rearranging for  $y$ :

$$y = \frac{I}{p_y} - \frac{p_x}{p_y}x$$

- **Intercept:**  $I/p_y$  (max amount of  $y$  if  $x = 0$ )
- **Slope:**  $-p_x/p_y$  (opportunity cost of  $x$  in terms of  $y$ )



# Budget Constraint Graph



Budget Line with  $I = 100$ ,  $p_x = 10$ ,  $p_y = 5$

# The Consumer's Problem

The consumer chooses  $(x, y)$  to:

$$\max_{x,y} U(x, y)$$

subject to:

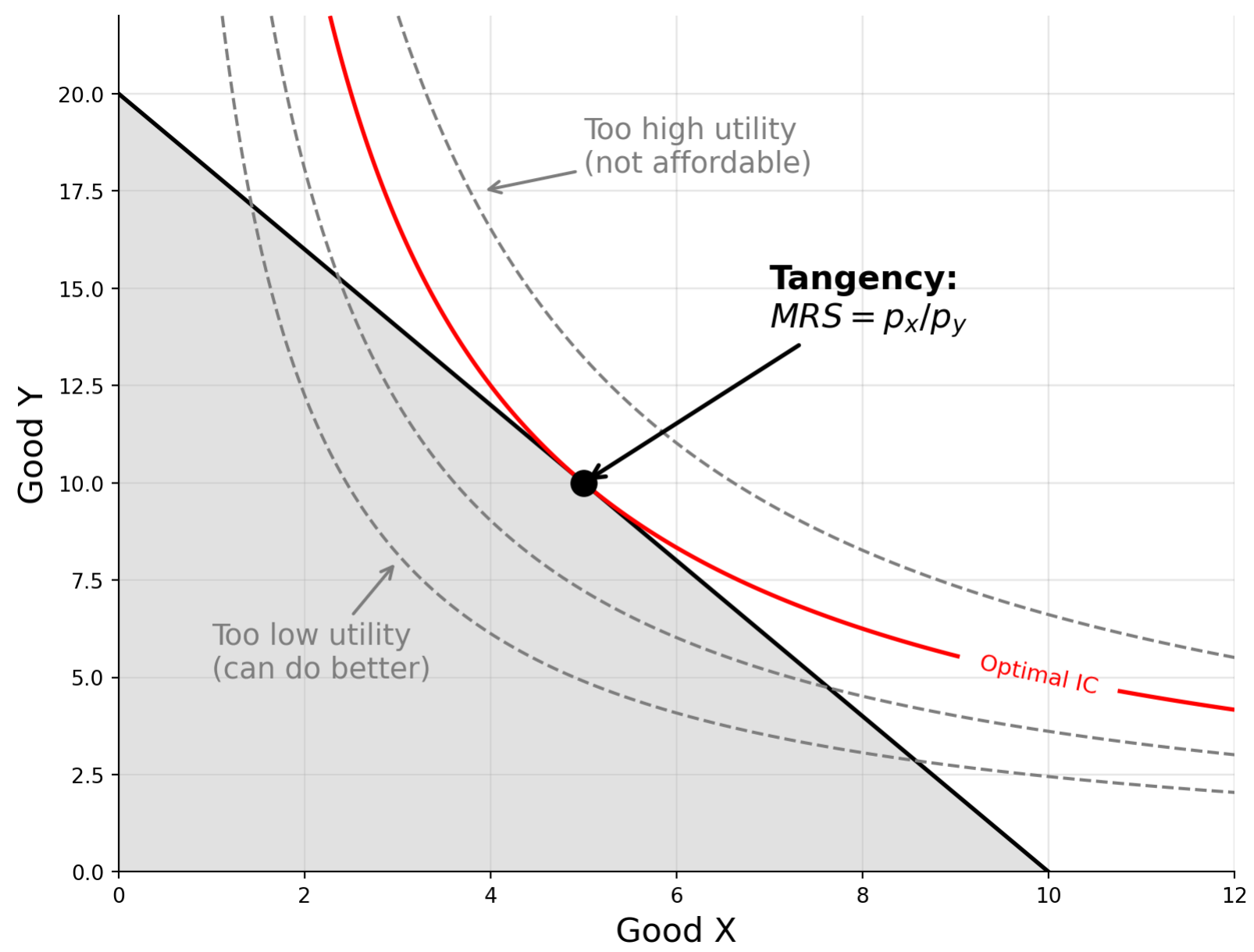
$$p_x x + p_y y = I$$

$$x \geq 0, \quad y \geq 0$$

**Goal:** Find the highest indifference curve that touches the budget line.

**Intuition:** Get as much utility as possible given your budget.

# Graphical Solution



# Why Tangency is Optimal

At the tangency point:  $MRS = p_x/p_y$

**Intuition:** Consumer's subjective tradeoff (MRS) equals market tradeoff

- **MRS:** How much  $Y$  you're willing to give up for 1 unit of  $X$
- $p_x/p_y$ : How much  $Y$  you must give up (in market) for 1 unit of  $X$

If  $MRS > p_x/p_y$ :

- You value  $X$  more than market does
- Should buy more  $X$ , less  $Y$

If  $MRS < p_x/p_y$ :

- You value  $X$  less than market does
- Should buy less  $X$ , more  $Y$

# The Lagrangian Method

The consumer's problem:

$$\max_{x,y} U(x,y) \quad \text{subject to} \quad p_x x + p_y y = I$$

Denote  $\lambda$  as the **Lagrange multiplier** and setup the **Lagrangian**:

$$L(x, y, \lambda) = U(x, y) + \lambda(I - p_x x - p_y y)$$

**First-order conditions** (FOCs):

1.  $\frac{\partial L}{\partial x} = \frac{\partial U}{\partial x} - \lambda p_x = 0$
2.  $\frac{\partial L}{\partial y} = \frac{\partial U}{\partial y} - \lambda p_y = 0$
3.  $\frac{\partial L}{\partial \lambda} = I - p_x x - p_y y = 0$

# First-Order Conditions

From the first two FOCs:

$$MU_x = \lambda p_x \quad \text{and} \quad MU_y = \lambda p_y$$

Dividing these:

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y}$$

This is exactly the tangency condition: **MRS = price ratio!**

Solving the system of equations given by the FOCs yields the optimal consumption bundle  $(x^*, y^*)$  and the multiplier  $\lambda^*$ .

# Interpretation of $\lambda$

From FOCs:  $\lambda = \frac{MU_x}{p_x} = \frac{MU_y}{p_y}$

**$\lambda$  = marginal utility of income**

- How much utility increases if income increases by \$1
- Measures the “value” of relaxing the budget constraint
- Important for welfare analysis

**Example:** If  $\lambda = 0.5$ :

- \$1 more of income increases utility by 0.5 utils
- Equivalently: consumer willing to pay \$2 for 1 more util

**Note:**  $\lambda$  decreases as income increases (diminishing marginal utility of income)

# Example: Cobb-Douglas Utility

**Setup:**  $U(x, y) = x^\alpha y^\beta$ , budget:  $p_x x + p_y y = I$

**Step 1:** Form the Lagrangian

$$L = x^\alpha y^\beta + \lambda(I - p_x x - p_y y)$$

**Step 2:** Take FOCs

$$\frac{\partial L}{\partial x} = \alpha x^{\alpha-1} y^\beta - \lambda p_x = 0$$

$$\frac{\partial L}{\partial y} = \beta x^\alpha y^{\beta-1} - \lambda p_y = 0$$

$$\frac{\partial L}{\partial \lambda} = I - p_x x - p_y y = 0$$



# Example: Cobb-Douglas Utility (cont.)

**Step 3:** Solve the system of equations to find  $x^*$ ,  $y^*$ , and  $\lambda^*$ .

$$x^* = \frac{\alpha I}{(\alpha + \beta)p_x}, \quad y^* = \frac{\beta I}{(\alpha + \beta)p_y}$$

*See handout for full derivation.*

## Key results:

- **Expenditure on X:**  $p_x x^* = \frac{\alpha I}{\alpha + \beta}$
- **Expenditure on Y:**  $p_y y^* = \frac{\beta I}{\alpha + \beta}$
- **Expenditure shares are constant:**  $\frac{\alpha}{\alpha + \beta}$  and  $\frac{\beta}{\alpha + \beta}$

# Constant Expenditure Shares

**Cobb-Douglas utility implies constant expenditure shares regardless of income or prices.**

Which of the following goods do you think have roughly constant expenditure shares in real life?

1. Food
2. Housing
3. Travel
4. Charitable giving

**Engel's Law:** As income rises, the proportion spent on food decreases.

**How to model this?**

# **Application: Cash vs In-Kind Transfers**

# Food Stamps

**Policy question:** Give \$100 cash or \$100 food stamps?

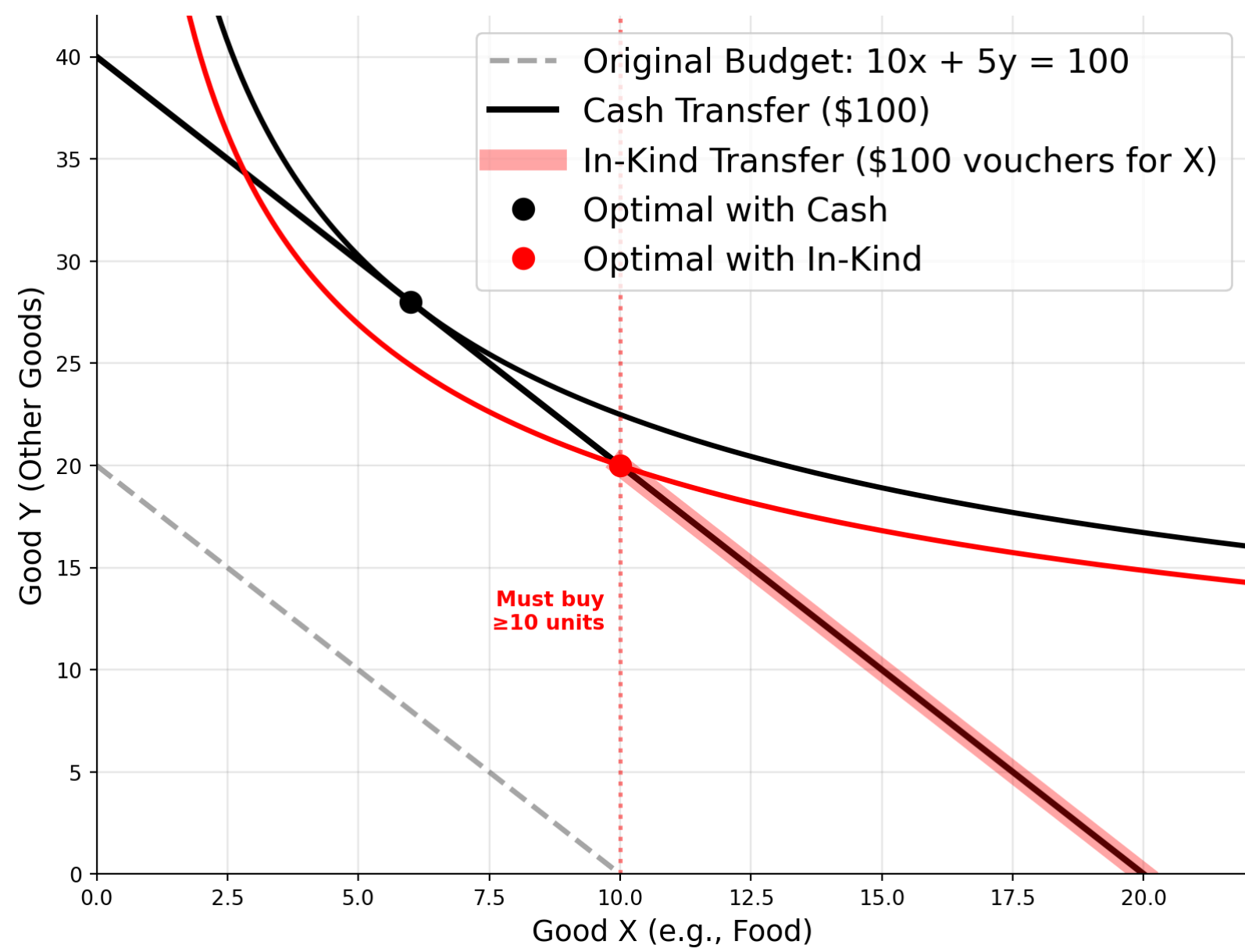
**Setup:**

- Two goods: food ( $x$ ) and other goods ( $y$ )
- Original income:  $I$

**With cash:** Budget is  $(p_x, p_y, I + 100)$

**With food stamps:** Can buy up to  $100/p_x$  extra food, but must spend at least that on food

# Cash vs Food Stamps on a Graph



# Cash vs In-Kind Transfers

**Key insight:** Cash transfers are at least as good as in-kind transfers (eg. SNAP, housing vouchers, energy assistance etc.)

- More flexibility → reach higher indifference curve (or same)
- If in-kind doesn't bind, cash and in-kind are equivalent
- If in-kind binds, cash is strictly better

## Then why use in-kind transfers?

- Paternalism (society values food/housing more than recipients)
- Political economy (easier to justify to voters)
- Externalities (e.g., nutrition benefits for children)

# Summary

## What we covered:

1. **Preferences:** Axioms of rational choice, utility representation
2. **Indifference curves:** Properties, MRS
3. **Common utility functions:** Perfect substitutes, perfect complements, Cobb-Douglas, CES
4. **Utility maximization:** Tangency condition ( $MRS = p_x/p_y$ ), Lagrange method, Indirect utility
5. **Application:** Cash vs in-kind transfers